Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 3 only
(c) 1 and 4 only (d) 2, 3 and 4 only

Ans. (c) : Sugarcane is one of the important commercial crops of India. It is a tropical and subtropical crop. Sugarcane crop is grown under 21ºC to 27ºC temperature and annual rainfall of 75 cm to 100 cm with humid climate is all it required. In the nursery method of sowing sugarcane, the bud chip method can produce on an average 20 plants from a sowed sugarcane ; by this method, about 80 percent of the seed is saved. Sugarcane cultivation can be done with a setting prepared with tissue culture.

49. In the context of India, which of the following is/are considered to be practice(s) of eco-friendly agriculture?

1. Crop diversification
2. Legume intensification
3. Tensiometer use
4. Vertical farming

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 3 only
(c) 4 only (d) 1, 2, 3 and 4

Ans. (a) : Crop diversification refers to the addition of new crops or cropping systems to agricultural production on a particular farm, taking into account the different returns from value-added crops with complimentary marketing opportunities. Introducing a greater range of vanishes in a particular agro-ecosystem leads to diversification of agricultural production, which can also increase natural biodiversity, strengthening the ability of the agro-ecosystem to respond to these stresses. So statement 1 is correct.

The discourse on legume intensification is currently framed as sustainable intensification (SI) and commonly encompasses three dimensions: (1) Increased productivity, (2) Maintenance of ecosystem services and (3) Increased resilience to shocks. So 2 is correct.

When the pressure in the tensiometer is determined to be in equilibrium with the water pressure in the soil, the tension meter gauge reading represents the matric potential of the soil. Such tension meters are used in irrigation scheduling to help farmers and other irrigation managers to determine when to water. It is also considered an eco-friendly agriculture as it helps control the excessive use of water, distorting the quality of the soil. So, statement 3 is correct.

50. What is the use of biochar in farming?

1. Biochar can be used as a part of the growing medium in vertical farming.
2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for longer time.

Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Biochar has the potential to produce farm - based renewable energy in an eco-friendly way. Specifically, the quality of biochar depends on several factors, such as the type of soil, metal and the raw material used for carbonization, the pyrolysis conditions and the amount of biochar applied to the soil.

In addition, the biochar amendment to the soil proved to be beneficial to improve enhancing plant growth, since biochar contains organic matter and nutrients, its addition increased soil pH, electric conductivity (EC), organic carbon (C), total nitrogen (N), available phosphorous (P) and the cation exchange capacity (CEC). Earlier, it reported that the biochar application affected the toxicity, transport of various heavy metals in the soil due to improved soil absorption capacity. The presence of plant nutrients and ash in the biochar and its large surface area, porous nature and the ability to act as a medium for microorganisms have been identified as the main reasons for the improvement in soil properties and increase in the absorption of nutrients by plants in soils treated with biochar.

Biochar application decreased the tensile strength of soil cores, indicating that the use of biochar can reduce the risk of soil compaction. A lot has already been discussed on the benefits of inoculation of rhizobacteria in soil, but the addition of biochar can also provide more nutrients to the soil, thus benefitting the agricultural crops. The mixing of the plant growth-promoting microorganisms with biochar was referred to as the best combination for the growth and yield of french beans. Therefore, all statements are correct.

51. In India, which of the following can be considered as public investment in agriculture?

1. Fixing Minimum Support Price for agricultural produce of all crops.
2. Computerization of Primary Agricultural Credit Societies.
3. Social Capital development.
4. Free electricity-support to farmers.
5. Waiver of agricultural loans by the banking system.
6. Setting up of cold storage facilities by the governments.

Select the correct answer using the code given below:

(a) 1, 2 and 5 only (b) 1, 3 4 and 5 only
(c) 2, 3 and 6 only (d) 1, 2, 3, 4, 5 and 6

Ans. (c) : Public investment in agriculture constitutes investment made by the government to create capital or social assets to improve agricultural productivity. Fixing MSP gives farmers the remunerative price without creating any durables infrastructure. So, statement 1 is not correct.

Computerization Agri Credit Societies improve their productivity by digitising their functioning. Social capital development like Agri education improves the knowledge base of farmers, paving the way for improved productivity. So, statement 2 and 3 are correct.

While free electricity and waiver of Agri loans create government liabilities without creating any durable assets on the ground. So, statements 4 and 5 are not correct.

Cold chain infra also helps reduce post- harvest losses creating durable assets. So, statement 6 is correct.

52. Which of the following factors/policies were affecting the price of rice in India in the recent past?

1. Minimum Support Price
2. Government's trading
3. Government's stockpiling
4. Consumer subsidies

Select the correct answer using the code given below:

(a) 1, 2 and 4 only (b) 1, 3 and 4 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

Ans. (d) : By the following ways, the price of rice in India gets affected:

- MSP increases the overall price of rice in the market as the traders have to buy near MSP value otherwise, farmers have the option to sell to an appointed government agency at MSP.
- Government also carries out auctions of the excess stockpile or intervenes when market prices soar to stabilize them.
- Similarly, stockpiling is done to offload the stock when market price soar providing cheap rice to consumers and stocking when prices fall ensuring, remunerative prices to farmers.
- The government also provides subsidised food to the poor at fair price shops. This reduces the overall demand in the open market and prices are affected.

So, all the statements are correct.

53. Under the Kisan Credit Card scheme, short-term credit support is given to farmers for which of the following purposes?

1. Working capital for maintenance of farm assets.
2. Purchase of combine harvesters, tractors and mini trucks.
3. Consumption requirements of farm households.
4. Post-harvest expenses.
5. Construction of family house and setting up of village cold storage facility.

Select the correct answer using the code given below:

(a) 1, 2 and 5 only (b) 1, 3 and 4 only
(c) 2, 3, 4 and 5 only (d) 1, 2, 3, 4 and 5

Ans. (b) : The schemes aim at providing adequate and timely credit for the comprehensive credit requirements of farmers under the single window for their cultivation and other needs as indicated below:

- To meet the short term credit requirements for cultivation of crops.
- Post- harvest expenses.
- Produce marketing loan.
- Consumption requirement of farmer household.
- Working capital for maintenance of farm assets, activities allied to agriculture, like dairy animals inland fishery and also working capital required for floriculture, horticulture etc.
- Investment credit requirement for agriculture and allied activities like pump sets, sprayers, dairy animals, floriculture, horticulture etc.
- Short term credit requirements of rearing of animals, birds, fish, shrimp, other aquatic organism, capture of fish.

54. Consider the following statements:

1. In the case of all cereals, pulses and oil-seeds, the procurement at Minimum Support Price (MSP) is unlimited in any State/UT of India.
2. In the case of cereals and pulses, the MSP is fixed in any State/UT at a level to which the market price will never rise.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : The procurement of oilseeds, pulses and cereals is not unlimited. Under PM-AASHA guidelines, limit the procurement to 25% of pulses and oil seeds production at MSP. So, statement 1 is not correct.

Market price, by definition, is determined through demand and supply forces. So it is difficult to predict whether the market price for any crop must be lower or higher as compared to MSP. So, statement 2 is also not correct. Hence, the correct answer is (d).